A singleton end forces a further unique crossing

Human Compression project: L044 continuation

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le A_n$, $n \ge 12$, be transitive with a connected string C-marking. Assume every maximal ambient omission G_j is intransitive and retain the smaller-degree alternating induction of the complete critical-tail reduction. Suppose G_0 has orbits $\{a\}$ and $B = \Omega \setminus \{a\}$. Put $b = a^{\rho_0}$ and $H = G_{\ge 2}$. An eligible edge always crosses *ambient* G_j-orbits.

Theorem 1. *If every colour $j \ge 2$ has at least two eligible edges, then $r \le \lfloor (n-1)/2 \rfloor$. Consequently any counterexample with this singleton end has a further colour $j \ge 2$ with exactly one ambient-eligible edge in the whole permutation graph.*

Proof. Suppose the rank bound fails. The accepted critical odd-tail reduction gives $n = 2r \ge 14$, $G_0 = A_B$ naturally, and H intransitive on its support. The group H fixes a, b: it fixes a and commutes with ρ_0. Write $f = |\operatorname{Fix}(H)|$ and let c count its nontrivial orbits. The full ambient interval theorem, using the two eligible edges of every colour in the actual interval $\{2, \dots, r-1\}$, gives

$$2(r-2) \le n - f - c, \qquad f + c \le 4.$$

Since $f \ge 2$ and $c \ge 2$, necessarily $f = c = 2$. Thus $X = \Omega \setminus \{a, b\}$ is the union of exactly two nontrivial H-orbits.

Write $\rho_0 = (a\ b)\delta$, where δ is an odd involution on X centralizing H. If δ has no fixed point on X, the complete accepted endpoint-centralizer theorem already gives the rank bound. Its hypotheses are exactly the present transitive even string C-action with intransitive maximal omissions and singleton G_0-orbit. That theorem does not assume that H is transitive on X; its full actual-orbit and paired-row arguments remain paid here.

We may therefore assume that δ has a fixed point in X. Commutation permutes actual H-orbits. It cannot exchange the two orbits, since that would make it fixed-point-free. On either preserved orbit a centralizing permutation fixing one point fixes every point. Thus δ is the identity on one orbit and acts freely on the other; it cannot be the identity on both because it is odd. Denote these orbits by V and O, respectively.

For any colour $j \ge 2$, an eligible j-edge in O is carried by ρ_0 to another eligible j-edge in O: the generators commute, and $\rho_0 \in G_j$ preserves each ambient omission orbit. These two edges are distinct. Indeed a fixed unordered edge of the fixed-point-free involution δ would itself be a δ-pair, and its endpoints would already be joined by $\rho_0 \in G_j$, contradicting eligibility. If there is no eligible j-edge in O, its two assumed eligible edges both lie in V, since H fixes a, b. Every colour therefore has an eligible pair in one actual H-orbit. The complete paired-head theorem applies and gives the contradiction.

Finally every colour has at least one eligible edge: if all its edges stayed within omission orbits, adjoining its generator could not turn the intransitive omission into transitive G. The negation of the theorem's paired hypothesis consequently gives exactly one eligible edge for some $j \ge 2$, as stated. □

Scope and attribution. This is a new routing consequence for the singleton continuation of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Propositions 5.31–5.32, physical pages 42–45. It retains the complete critical-tail, ambient-interval, endpoint-centralizer and paired-head proofs, including their classical inputs and smaller-degree induction. The new step converts global pairs to a pair in one actual orbit using the exact

two-orbit count and the fixed-point-free complement theorem. No projected IP or local-to-ambient eligibility is inferred. The further unique crossing and other full-source obligations still require resolution. No new computation, formalization, original or question credit is claimed.